

MATH 345 Skeleton Notes

Unit 2: Anatomy of a linear map

Section 2.5: Inverses and determinants

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Matrix invertibility and linear systems

In this section we finish the square-matrix version of the Unit 2 story. A square matrix is invertible exactly when it “does not forget any nonzero input direction” and “every output is reached once”. The inverse matrix gives the inverse map when it exists. The determinant, a fundamental quantity in linear algebra that we only briefly discuss in this course, gives a single-number test for invertibility.

Definition: Invertibility of Matrices An $n \times n$ matrix A is called nonsingular, or invertible, if there exists an $n \times n$ matrix B such that $AB = BA = I_n$. Such a B is called the inverse of A . If no such B exists, A is called singular or noninvertible.

Note The inverse of a matrix mimics the reciprocal of a real number.

Activity

Let

$$A = \begin{bmatrix} 1 & 3 \\ 1 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 4 & -3 \\ -1 & 1 \end{bmatrix}.$$

Compute both AB and BA , and make a conclusion using the language of inverses.

Fact The inverse of a matrix, if it exists, is unique. Therefore, we can write A^{-1} for the inverse of A .

Fact For $n \times n$ matrices A and B , if $AB = I_n$, then $BA = I_n$.

Activity

Does $A = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$ have an inverse?

Theorem If both A and B are nonsingular $n \times n$ matrices, then the matrix AB is nonsingular and its inverse is $(AB)^{-1} = B^{-1}A^{-1}$.

Proof It suffices to calculate that $(AB)(B^{-1}A^{-1}) = I_n$ and $(B^{-1}A^{-1})(AB) = I_n$. Firstly,

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = AA^{-1} = I_n.$$

Similarly,

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_nB = B^{-1}B = I_n.$$

Fact: Follow Up Facts About Invertibility

- If $A_1, A_2, \dots, A_k$ are $n \times n$ invertible / nonsingular matrices, then $A_1A_2 \cdots A_k$ is invertible / nonsingular and

$$(A_1A_2 \cdots A_k)^{-1} = A_k^{-1} \cdots A_2^{-1}A_1^{-1}.$$

- If A is invertible/nonsingular, then A^{-1} is invertible/nonsingular and

$$(A^{-1})^{-1} = A.$$

- If A is invertible/nonsingular, then A^T is invertible/nonsingular and

$$(A^{-1})^T = (A^T)^{-1}.$$

Let $T = T_A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a matrix transformation induced by the $n \times n$ matrix A . What does invertibility imply about the geometry of the transformation T_A ?

Consider $T' = T_{A^{-1}}$. What happens when we compose them?

Activity

Compute the composition of T and T' in both orders.

Definition Given a matrix transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$, if there exists a transformation $S : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $S(T(\mathbf{x})) = T(S(\mathbf{x})) = \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$, we say that T has an inverse.

Theorem Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ denote the matrix transformation induced by an $n \times n$ matrix A . Then:

1. A is invertible if and only if T has an inverse.
2. In this case, T has exactly one inverse (which we denote as T^{-1}), and $T^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the transformation induced by the matrix A^{-1} .

In other words, $(T_A)^{-1} = T_{A^{-1}}$.

T and T^{-1} ‘reverse’ or ‘undo’ the action of each other.

Activity

Consider $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by reflection across the x -axis. Find the matrix representing this transformation, and find the inverse.

Theorem The 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is invertible if and only if $ad - bc \neq 0$, in which case

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

We define the **determinant** of the matrix A to be the quantity $\det(A) = ad - bc$. Thus the matrix A is invertible if and only if its determinant is nonzero.

For now, $\det(A) = ad - bc$ is only the determinant of a 2×2 matrix. We will define determinants for larger square matrices later in this section.

Activity: Reading determinant checks in code

The following code computes determinants of two 2×2 matrices.

```
import numpy as np

A = np.array([
    [1., 1.],
    [0., 1.],
])

B = np.array([
    [1., 0.],
    [0., 0.],
])

detA = float(np.linalg.det(A))
detB = float(np.linalg.det(B))

detA, detB
```

Output:

```
(1.0, 0.0)
```

1. Which matrix is invertible?
2. Which matrix forgets a nonzero input direction?
3. How does the determinant tell you?

Activity

Find the inverse of each matrix, if possible.

$$A = \begin{bmatrix} -2 & -3 \\ 4 & 6 \end{bmatrix}$$

Activity

Find the inverse of each matrix, if possible.

$$B = \begin{bmatrix} 1 & 6 \\ 2 & 3 \end{bmatrix}$$

If A is an $n \times n$ matrix, then the linear system $A\mathbf{x} = \mathbf{b}$ is a system of n equations in n unknowns. Suppose A is nonsingular. How can we use A^{-1} to solve the system $A\mathbf{x} = \mathbf{b}$?

Activity

Solve the system using the inverse of A .

Consequences:

- When A^{-1} exists, then $A\mathbf{x} = \mathbf{b}$ has a unique solution.
- If A is invertible/nonsingular, then the *only* solution to the homogeneous system $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$.
- Put differently, an invertible square matrix forgets no nonzero input direction.

Activity

Suppose that

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Use this information to calculate the following:

$A\mathbf{x} = \mathbf{b}$

Activity

Suppose that

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Use this information to calculate the following:

$$A\mathbf{x} = \mathbf{c}$$

Activity

Suppose that

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Use this information to calculate the following:

$$A\mathbf{x} = \mathbf{0}$$

Activity

Consider the transformation $P : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$P\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) = \begin{bmatrix} a_1 \\ 0 \end{bmatrix}.$$

Is P a matrix transformation? If so, find the matrix A corresponding to P .

Activity

Consider the transformation $P : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$P \left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \right) = \begin{bmatrix} a_1 \\ 0 \end{bmatrix}.$$

Is P an invertible transformation?

Calculating inverses

Consider vectors $\mathbf{c}_1, \dots, \mathbf{c}_n$ in $\mathbb{R}^m$, and let A be the $m \times n$ matrix whose columns are the vectors $\mathbf{c}_1, \dots, \mathbf{c}_n$.

Note that $\{\mathbf{c}_1, \dots, \mathbf{c}_n\}$ are linearly independent precisely when the equation $A\mathbf{x} = \mathbf{0}$ has *only* the *trivial* solution.

Note that additionally, $\text{span}\{\mathbf{c}_1, \dots, \mathbf{c}_n\} = \mathbb{R}^m$ exactly when $A\mathbf{x} = \mathbf{b}$ has a solution for all $\mathbf{b}$ in $\mathbb{R}^m$.

We use these facts to extend our invertibility criteria for matrices.

Fact: Invertibility Criteria Let A be a $n \times n$ matrix. Then the following are equivalent:

- A is invertible.
- The columns of A are linearly independent.
- The columns of A span $\mathbb{R}^n$.
- The rows of A are linearly independent.
- The rows of A span $\mathbb{R}^n$.

Activity

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 3 \\ 0 \\ -5 \end{bmatrix}.$$

Is S linearly dependent?

For a given $n \times n$ matrix A , if we are looking for an inverse for A (call this matrix X), we must have $AX = I_n$. We are solving n linear systems simultaneously:

Activity

Describe the system of linear equations for finding the inverse.

Since, to solve each system, we wish to reduce the matrix A down to its RREF each time, we can combine all the systems into one big augmented matrix:

Activity

Write the augmented matrix for finding the inverse.

It follows from the theorem that if $[A \mid I_n]$ is row reducible to $[B \mid C]$, then $BX = C$. In particular, if we can row reduce $[A \mid I_n]$ to $[I_n \mid C]$, then it follows that $X = C$. Gaussian elimination (The Gaussian Elimination Algorithm) will reduce a given matrix to this form, provided that an inverse exists.

1. Set up the augmented matrix $[A \mid I_n]$.
2. Apply elementary row operations to $[A \mid I_n]$ to reduce it to $[I_n \mid A^{-1}]$.

If A is an $n \times n$ matrix, either A can be reduced to I_n by elementary row operations or it cannot. In the first case, the algorithm produces A^{-1} ; in the second case, A^{-1} does not exist.

Activity

Find A^{-1} where $A = \begin{bmatrix} 1 & 1 & 4 \\ 2 & 3 & 2 \\ 0 & 0 & 1 \end{bmatrix}$.

Theorem: Box of Facts/Invertibility Criteria Expanding on Invertibility Criteria, we can add row-reduction and solution-set criteria to the same list.

Let A be an $n \times n$ matrix. The following are equivalent:

1. A is nonsingular/invertible.
2. The columns of A are linearly independent.
3. The columns of A span $\mathbb{R}^n$.
4. The rows of A are linearly independent.
5. The rows of A span $\mathbb{R}^n$.
6. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
7. A is row equivalent to I_n .
8. The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ matrix $\mathbf{b}$.
9. The rank of A is n .

General determinants

In this section we consider square $n \times n$ matrices. Our *goal* is to find a number, called the *determinant* that we can calculate for any matrix, which tells us whether or not the matrix is invertible.

Let's start with the 2×2 case. Recall that in the theorem we defined the determinant of a 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

to be $\det(A) = ad - bc$. It has the required properties of the determinant that we will now define for more general matrices.

Activity

Find the determinant of the following matrices:

$$A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$$

Activity

Find the determinant of the following matrices:

$$B = \begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$$

We use the 2×2 determinant to define the 3×3 determinant. The idea is recursive: a 3×3 determinant is built from 2×2 determinants. Later, an $n \times n$ determinant will be built from $(n - 1) \times (n - 1)$ determinants.

Definition: Minors of a 3×3 matrix Let A be a 3×3 matrix. The **(i, j) -th minor** of A , denoted

$$\text{minor}(A)_{ij},$$

is the determinant of the 2×2 matrix obtained from A by deleting row i and column j .

Activity

Suppose

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}.$$

Calculate $\text{minor}(A)_{11}$.

Activity

Suppose

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}.$$

Calculate $\text{minor}(A)_{23}$.

Definition: Cofactors of a 3×3 matrix Let A be a 3×3 matrix. The

(i, j) -cofactor

of A , denoted

$$\text{cof}(A)_{ij},$$

is

$$\text{cof}(A)_{ij} = (-1)^{i+j} \text{minor}(A)_{ij}.$$

Activity

Define

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

Compute $\text{cof}(A)_{11}$.

Activity

Define

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

Compute $\text{cof}(A)_{23}$.

Definition: Determinant of a 3×3 matrix Let $A = [a_{ij}]$ be a 3×3 matrix. The **determinant** of A , denoted $\det(A)$, is defined by expansion along the first row:

$$\det(A) = a_{11} \text{cof}(A)_{11} + a_{12} \text{cof}(A)_{12} + a_{13} \text{cof}(A)_{13}.$$

This process is called

cofactor expansion.

Fact For a 3×3 matrix, cofactor expansion along any row or any column gives the same determinant.

Activity

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}.$$

Find the determinant of A by expanding along row 1.

Activity

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}.$$

Find the determinant of A by expanding along column 2.

Note To track the sign $(-1)^{i+j}$ when computing cofactors, it may be helpful to view the matrix as a checkerboard as below:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

The sign in the i th row and j th position will be the sign of $(-1)^{i+j}$.

Note Since any row or column gives the same determinant, choose a row or column with many zeros when possible.

Activity

Compute the determinant of the following matrices:

$$A = \begin{bmatrix} 3 & -5 & 1 \\ 0 & 1 & 0 \\ -2 & -4 & 7 \end{bmatrix}$$

Activity

Compute the determinant of the following matrices:

$$B = \begin{bmatrix} 0 & \pi & 200 \\ 0 & 0.03 & 4 \\ 0 & 19 & -3584 \end{bmatrix}$$

Note Whenever a row or column of a matrix only has zeros as entries, expanding along that row shows that the determinant of the matrix is always zero, so no cofactor computations are required.

Now that 3×3 determinants have been defined, we can define larger determinants recursively. For a 4×4 matrix, each minor determinant is a 3×3 determinant. For a 5×5 matrix, each minor determinant is a 4×4 determinant, and so on.

Definition: Determinant of a general square matrix Let $A = [a_{ij}]$ be an $n \times n$ matrix with $n \geq 4$. Assume determinants have already been defined for $(n - 1) \times (n - 1)$ matrices. The (i, j) -th minor of A , denoted

$$\text{minor}(A)_{ij},$$

is the determinant of the $(n - 1) \times (n - 1)$ matrix obtained from A by deleting row i and column j .

The (i, j) -cofactor of A is

$$\text{cof}(A)_{ij} = (-1)^{i+j} \text{minor}(A)_{ij}.$$

The determinant of A is defined by expansion along the first row:

$$\det(A) = \sum_{j=1}^n a_{1j} \text{cof}(A)_{1j}.$$

Fact For an $n \times n$ matrix, cofactor expansion along any row or any column gives the same determinant:

$$\det(A) = \sum_{j=1}^n a_{ij} \text{cof}(A)_{ij}$$

for any fixed row i , and

$$\det(A) = \sum_{i=1}^n a_{ij} \text{cof}(A)_{ij}$$

for any fixed column j .

Activity

Calculate the determinant of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 4 & 2 & 3 \\ 1 & 3 & 4 & 5 \\ 3 & 4 & 3 & 2 \end{bmatrix}$$

Activity

Compute $\det(A)$, where

$$A = \begin{bmatrix} 3 & 2 & 3 & 4 \\ 0 & 4 & 2 & 3 \\ 0 & 0 & 2 & 5 \\ 0 & 0 & 0 & -1 \end{bmatrix}.$$

Recall Main diagonal.

Definition: Triangular and Diagonal Matrices An $n \times n$ matrix $A = [a_{ij}]$ is called

upper triangular

if $a_{ij} = 0$ for $i > j$, and

lower triangular

if $a_{ij} = 0$ for $i < j$. A matrix is

diagonal

if it is both upper triangular and lower triangular, i.e., if $a_{ij} = 0$ for $i \neq j$. We denote the $n \times n$ diagonal matrix with entries $\lambda_1, \dots, \lambda_n$ on the diagonal by $\text{diag}(\lambda_1, \dots, \lambda_n)$.

Theorem If $A = [a_{ij}]$ is a upper or lower triangular matrix, then $\det(A) = a_{11} \cdots a_{nn}$. That is, the determinant of a triangular matrix is the product of entries in the *main diagonal*.

Proof A generalization of the argument used to calculate the activity, i.e., expanding the determinant in cofactors along the first column iteratively.

Theorem If a row or column of a matrix A consists entirely of zeros, then $\det(A) = 0$.

Proof A generalization of the argument used to calculate the determinant in the activity.

Theorem If a matrix B is obtained from a matrix A by swapping two rows or two columns, then $\det(B) = -\det(A)$.

Activity

Check directly that the theorem is true when A and B are 2×2 matrices.

Proof If

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

and

$$B = \begin{bmatrix} c & d \\ a & b \end{bmatrix},$$

then $\det(A) = ad - bc$ and $\det(B) = cb - ad = -(ad - bc) = -\det(A)$.

Theorem If two rows (or columns) of A are equal to one another, then $\det(A) = 0$.

Proof Let B be the matrix obtained by swapping the two identical rows / columns of A . Then $A = B$, and the theorem implies that $\det(A) = -\det(B) = -\det(A)$. Therefore $2 \det(A) = 0$, and thus that $\det(A) = 0$.

Theorem If B is obtained from A by multiplying each entry in a row or column of A by a quantity k , then $\det(B) = k \det(A)$.

Proof Suppose that B is obtained from A by multiplying each entry of row i by k . Consider a cofactor expansion along row i of B . Then

$$\begin{aligned} \det(B) &= ka_{i1} \operatorname{cof}(B)_{i1} + ka_{i2} \operatorname{cof}(B)_{i2} + \cdots + ka_{in} \operatorname{cof}(B)_{in} \\ &= k[a_{i1} \operatorname{cof}(A)_{i1} + \cdots + a_{in} \operatorname{cof}(A)_{in}] \\ &= k \det(A). \end{aligned}$$

Fact If A is a $n \times n$ matrix, then $\det(kA) = k^n \det(A)$.

Theorem If B is obtained by adding k times the i th row of A to the j th row, for $i \neq j$, which is a row operation which we denote by $R_j \rightarrow R_j + kR_i$, then $\det(B) = \det(A)$.

We can check the theorem in the 2×2 case. If

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

and

$$B = \begin{bmatrix} a & b \\ c + ka & d + kb \end{bmatrix},$$

then

$$\det(A) = ad - bc$$

and

$$\begin{aligned} \det(B) &= a(d + kb) - b(c + ka) \\ &= ad + kab - bc - kab \\ &= ad - bc = \det(A). \end{aligned}$$

Theorem: How row operations affect the determinant Here is a summary of how the elementary row operations, when applied to a matrix, affect the determinant:

- $R_i \rightarrow kR_i$ multiplies the determinant by k .
- $R_i \leftrightarrow R_j$ multiplies the determinant by -1 .
- $R_j \rightarrow R_j + kR_i$ leaves the determinant unchanged.

Activity

Suppose a matrix B is obtained from a matrix A by applying the following row operations:

- $R_i \leftrightarrow R_j$
- $R_4 \rightarrow 4R_4$
- $R_3 \rightarrow R_3 + 3R_2$

If $\det(A) = 8$, what is the determinant of B .

Activity

Compute the determinant of the matrix

$$B = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 3 & 3 \\ 0 & 15 & 12 \end{bmatrix}.$$

Theorem If A and B are $n \times n$ matrices, then $\det(AB) = \det(A) \det(B)$.

Activity

Let

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 7 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}.$$

Compute the determinant of A , B , and C .

Activity

Let

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 7 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}.$$

Compute the determinant of AC and BC .

Theorem If A is a $n \times n$ matrix, then A is *nonsingular* if and only if $\det(A) \neq 0$. In this case $\det(A^{-1}) = 1/\det(A)$.

Proof We prove only one direction of the theorem, that if A is non-singular then $\det(A) \neq 0$ and $\det(A^{-1}) = 1/\det(A)$. Suppose that A is non-singular. Then $AA^{-1} = I$. Taking determinants of both sides, and using that I is diagonal, so that $\det(I) = 1$, we see that

$$1 = \det(I) = \det(AA^{-1}) = \det(A)\det(A^{-1}).$$

Thus $\det(A) \neq 0$ (since otherwise the right hand side of this equation would be zero), and rearranging gives $\det(A^{-1}) = 1/\det(A)$.

Theorem If A is a square matrix, then $\det(A) = \det(A^T)$.

Definition: Orthogonal Matrices A square matrix A is **orthogonal** if $A^T = A^{-1}$.

Activity

If A is an orthogonal matrix, what are the possible values of $\det(A)$?